Cryptocurrency Price Prediction Using LSTM Models and Media Sentiment Analysis

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Abstract

Cryptocurrencies are highly volatile assets influenced by complex factors, including social media sentiment. This project proposes an analytical framework that combines LSTM-based deep learning with media sentiment analysis to forecast cryptocurrency prices. We will formulate clear research questions and objectives. We will review recent literature to identify gaps and employ multiple data sources and model variants. We will integrate historical OHLCV price data with aggregated sentiment indicators from Twitter, Reddit, and news feeds. Our methodology covers data collection (using the Twitter and CryptoCompare APIs), feature engineering (including technical indicators and sentiment scores), and model architectures (LSTM, Bi-LSTM, and CNN-LSTM with attention). We will use hyperparameter tuning and robust baselines (ARIMA, XGBoost). Evaluation will use standard metrics (RMSE, MAPE, directional accuracy) and apply careful train/validation splits to avoid look-ahead bias[2, 3]. We address key challenges like data non-stationarity, sentiment noise, and model overfitting. Expected outcomes include novel insights into the role of sentiment in crypto markets and an open-source codebase.

1 Introduction

Cryptocurrency markets, such as those for Bitcoin and Ethereum, exhibit extreme volatility due to speculative trading and decentralized dynamics [8, 9]. This volatility creates both opportunities and risks for investors. It motivates the need for accurate price forecasting models. Traditional time-series methods often fail to capture such complex market behavior. Recent studies suggest that social media sentiment can serve as a leading indicator of market mood [6, 7]. For example, traders frequently discuss crypto on Twitter and Reddit. Sentiment metrics from these platforms have been shown to correlate with price moves, consistent with the Efficient Market Hypothesis[8].

Machine learning advances, especially deep learning sequence models (LSTM, GRU, BiLSTM), have improved forecasting by capturing long-term dependencies [1, 3]. Attention mechanisms and pretrained NLP models (BERT, RoBERTa) enable more sophisticated sentiment extraction [2, 5]. However, integrating qualitative sentiment data with quantitative price features remains challenging. This project addresses this gap by designing a hybrid predictive pipeline (see Fig. 1). The pipeline fuses historical price signals with media sentiment features. We will systematically explore model architectures, feature designs, and benchmark performance improvements from sentiment integration. The problem statement is: Can deep learning models combining price data and media sentiment significantly improve cryptocurrency price prediction compared to price-only models? We aim to answer this through rigorous experiments and analysis.

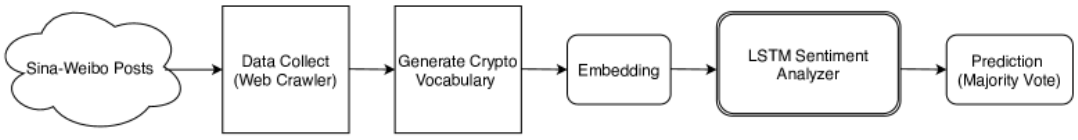


Figure 1: Architecture of LSTM-based cryptocurrency sentiment analysis and price prediction (adapted from Huang et al. 2021)[1].

Social media posts (Weibo/Twitter) are processed with a crypto-specific sentiment pipeline, producing sentiment scores that are fed into an LSTM network along with historical price data for predicting price movements.

1.1 Aims and Objectives

The primary research question is: How effectively can LSTM-based deep learning models incorporating media sentiment predict cryptocurrency prices? Specific objectives are:

- **Q1:** Identify which sentiment features (social media, news) contribute most to price forecast accuracy.
- **Q2:** Compare LSTM variants (standard LSTM, Bi-LSTM, CNN-LSTM, LSTM with attention) in this context.
- **Q3:** Evaluate how different data sources (Twitter, Reddit, news) and pre-trained sentiment models (VADER, FinBERT, RoBERTa) impact results.

- **Q4:** Establish robust baselines (ARIMA, XGBoost, classic RNNs) to quantify gains from sentiment and deep learning.
- **Q5:** Ensure reproducibility through open data/code, enabling future research and validation.

Key objectives include (1) collecting and curating multi-source datasets for crypto prices and sentiment, (2) engineering features such as lagged price indicators and sentiment momentum, (3) designing and training LSTM-based models with careful hyperparameter tuning, (4) evaluating performance using multiple metrics and ablation, and (5) documenting code and data for sharing. These objectives aim to advance understanding of the role of media sentiment in crypto forecasting. They also seek to deliver a validated prediction framework.

1.2 Related Work

Numerous studies have explored machine learning-based cryptocurrency forecasting. Early work focused solely on price time series; for example, several studies have compared statistical time-series baselines such as ARIMA against recurrent neural network models for Bitcoin. More recent research integrates sentiment analysis:

- **Price-plus-sentiment models:** Huang et al. (2021) developed a cryptocurrency-specific sentiment dictionary based on Chinese Weibo data and incorporated these sentiment measures with price information using an LSTM classifier, resulting in significant improvements over an AR(1) baseline[1]. Girsang and Stanley (2023) combined Binance price data with FinBERT-extracted Twitter sentiment scores and employed a hybrid LSTM-GRU network to forecast Ethereum and Solana prices. Their results indicate that models incorporating sentiment outperform price-only models in directional prediction accuracy [2]. These studies demonstrate the forecasting value of social media sentiment, but often limit their focus to short prediction windows or particular assets.
- **Deep model variants:** Seabe et al. (2025) systematically compare LSTM, GRU, Bi-LSTM, and a custom Temporal Attention Model in the context of Bitcoin, utilizing sentiment features derived from both VADER and RoBERTa. Their findings indicate that a Bi-LSTM enhanced with RoBERTa embeddings achieved the lowest MAPE (2.01%), while deep contextual embeddings outperformed lexicon-based sentiment features[3]. Likewise, Tiwari et al. (2025) introduce an attention-augmented CNN-LSTM architecture to process Twitter and Reddit data for sentiment classification, followed by subsequent price prediction using a GRU. This hybrid model produced MAE=0.0315 and MAPE=5.95% for Bitcoin, outperforming ARIMA[4]. These studies emphasize advanced neural architectures but often implement sentiment classification and price regression as separate, sequential stages.
- **Sentiment feature engineering:** Nezhad et al. (2026) emphasize sentiment aggregation. They analyze over 1.1M tweets with RoBERTa and apply PCA to combine features. Their method uses a decay-weighted sentiment time series. Using k-means to define price movement regimes, they found that the GRU model performs best because of its sentiment-based architecture [5]. This highlights a gap: how sentiment is quantified and aligned with price labels is as important as the forecasting model itself.

- **Baselines and critiques:** Other works examine hybrids such as LSTM+XGBoost and pure machine-learning baselines, including SVMs and random forests with sentiment features. Recent reviews note a lack of consensus on data splits and metrics, making cross-study comparison difficult.

Gaps: Existing literature is promising but uneven. Many studies focus on Bitcoin or a few altcoins. They often lack rigorous baselines and reproducibility. Most sentiment analyses are English-centric (except Huang's Chinese study). There is little inclusion of multi-language or non-English sources. Few works incorporate high-frequency or multi-modal data, such as images and forums. Long-term validation and concept-drift handling are rarely addressed. Our project aims to fill these gaps by systematically comparing models on multiple coins (e.g., BTC, ETH) and exploring diverse sentiment sources (Twitter, Reddit, news APIs). We will release code and datasets for transparency. We will also include explainability (e.g., SHAP) to interpret feature importances, as used in related forecasting studies.

Comparison Table of Key Studies:

Study	Year	Asset	Sentiment source	Model	Key finding
Huang et al. [1]	2021	(not specified, generic crypto)	Sina-Weibo (Chinese tweets)	LSTM (custom embedding)	+18.5% precision vs AR baseline
Girsang and Stanley [2]	2023	Ethereum, Solana	Twitter (FinBERT)	Hybrid LSTM-GRU	The sentiment-enhanced model outperforms baselines
Seabe et al. [3]	2025	Bitcoin	Twitter (VADER, RoBERTa)	LSTM, GRU, Bi-LSTM, TAM	Bi-LSTM + RoBERTa achieved MAPE 2.01%
Tiwari et al. [4]	2025	Multiple (portfolio)	Twitter, Reddit (CNN-LSTM)	CNN-LSTM + attention; GRU regressor	MAE=0.0315, MAPE=5.95% vs ARIMA
Nezhad et al. [5]	2026	Bitcoin (short-term)	Twitter (RoBERTa + PCA)	GRU	Performance gain mainly from sentiment design (not model)

2 Methodology

Data Sources: We will gather historical price data and sentiment data for selected cryptocurrencies. Price data (OHLCV) can come from CoinGecko, CryptoCompare, Binance

API, or Kaggle datasets. The data will cover at least 3–5 years up to the present (e.g., Jan 2018–Dec 2025). Sentiment sources include:

- **Twitter/X:** Using the Twitter API (Academic track or third-party tools like Twitter Scraper) to collect crypto-related tweets (by hashtag/keyword, e.g. "#Bitcoin", "$ETH").
- **Reddit:** Via Pushshift or Reddit API for posts/comments in crypto subreddits (r/CryptoCurrency, r/Bitcoin).
- **News APIs:** Aggregators like NewsAPI or Google News for headlines/articles about major coins. We will store data with timestamps, ensuring synchronization (e.g. daily alignment). Metadata (retweet count, user metrics) may be included.

Feature Engineering:

- **Price Features:** We will use raw prices and compute technical indicators, including moving averages, RSI, volatility, and momentum. We will also include log returns. To handle scale differences, we will normalize features using methods such as min-max or z-score normalization [2].
- **Sentiment Features:** We will process text using NLP tools. Options include lexicon-based tools (VADER [7]) and transformer-based models (FinBERT [2], RoBERTa). We will experiment with both to compare their effectiveness. For each day, we will aggregate tweet and post sentiment using mean, median, or weighted sum. Advanced features may include sentiment momentum (change over time), a rolling standard deviation for sentiment volatility, and PCA-combined sentiment factors, as in Nezhad et al. (2026) [5]. If relevant, we will consider non-English content (such as Chinese Weibo). Domain-specific lexicons may also be created as done by Huang et al. [1].
- **Temporal Alignment:** We will ensure sentiment features are properly lagged (e.g., using the previous day's average sentiment to predict the next day's price) to avoid look-ahead bias, which occurs when information unavailable at prediction time leaks into the data. We will conduct exploratory lag analysis to identify the optimal time offsets for sentiment features.

Model Architecture: We will implement several deep learning models:

1. **LSTM and GRU:** Stacked LSTM networks with varying depth (1–3 layers), and a comparison GRU.
2. **Bi-directional LSTM (Bi-LSTM):** to leverage information from both past and future (as in recent Bi-LSTM forecasting studies [3]).
3. **CNN-LSTM and CNN-BiLSTM:** Prepend a 1D-CNN to process sequential features (inspired by Tiwari et al. 2025[4]) which may capture short patterns before temporal modeling.
4. **Attention models:** Incorporate temporal attention layers or transformer-like encoders to weigh significant time steps [3, 4].
5. **Hybrid models:** LSTM outputs may be combined with tree-based learners such as XGBoost to compare sequence-based predictions with engineered-feature baselines.

All models will take as input a multivariate time sequence combining price and sentiment features. Outputs can be either price regressions or classifications of up/down movements.

Training and Validation: Data will be split chronologically, e.g., train on 2018–2023, validate on 2024, test on 2025, to simulate real forecasting. Rolling-window cross-validation can further assess stability (moving the train/test window). We will tune hyperparameters (learning rate, hidden units, batch size) using grid search or Bayesian optimization on validation sets. To avoid overfitting, techniques like dropout, early stopping, and regularization will be used. We will also compare baseline models: ARIMA (for benchmark), Random Forest/XGBoost on engineered features, and simple MLPs.

Evaluation Metrics: Forecast accuracy will be measured using standard regression metrics: RMSE (root mean square error), MAE (mean absolute error), and MAPE (mean absolute percentage error) [2, 3]. If framing a classification task (price up/down), we will use accuracy, F1-score, and directional accuracy. We will also report R^2 and, if appropriate, the Diebold-Mariano test for statistical significance between models. Performance will be compared both overall and in different market regimes (bull vs bear periods).

Reproducibility: All code will be developed in Python (e.g., TensorFlow/PyTorch) and shared via a public repository. Data processing scripts and trained model weights (if feasible) will be available. Where APIs are used (Twitter, news), we will log query parameters. Datasets will be described and, if licensing permits, posted (e.g., Google Drive/GitHub). We will provide a requirements.txt and instructions for replicating results, following best practices in ML reproducibility.

2.1 Project Challenges

Key challenges anticipated include the following:

- **Data Quality and Availability:** Crypto and social media data are often noisy and incomplete. We may face missing data, outliers (e.g., flash crashes), and API rate limits. Cleaning and imputation strategies will be required.
- **Non-Stationarity and Concept Drift:** The crypto market evolves rapidly; patterns in one year may not hold in another. Models must adapt to drift. We will consider retraining on sliding windows or online learning.
- **Sentiment Noise and Bias:** Social media sentiment can be skewed by spam, bots, or hype. Lexicon tools (VADER) may misinterpret slang or sarcasm. We may need to filter bot activity and validate sentiment signals.
- **Feature Alignment:** Ensuring price and sentiment data align temporally is non-trivial. We must carefully decide sentiment lag and aggregation windows to prevent leakage.
- **Overfitting:** With complex models and limited data (especially when many features are present), overfitting is a risk. We will monitor validation loss and use regularization. Ensemble or simpler models may sometimes outperform deep nets.
- **Interpretability:** Deep models are "black boxes". We plan to use explainability tools (e.g., SHAP values as in [8] or [3]) to quantify feature importance, which can also aid trust.
- **Ethics and Privacy:** Using social media data raises privacy issues. We will anonymize data as required and abide by the platform's terms. Also, model misuse (e.g. for market manipulation) is a concern, we will discuss ethical considerations in the report.

2.2 Project Team

The project team will work collaboratively across data collection, model development, evaluation, documentation, and reporting. Responsibilities are assigned to make each stage accountable while still allowing peer review and shared decision making.

1. **Nguyen Gia Long:** has a background in business administration, specializing in technology and innovation. He brings experience in marketing, communication, adaptability, and stakeholder facing work.
2. **Manoj Kumar:** has experience in software engineering, finance, and accounting. His skills include JavaScript, TypeScript, Python, HTML, CSS, ReactJS, MongoDB, SQL, REST APIs, Power BI, Excel Macros, AutoHotKey, data analysis, and data visualization.
3. **Md Aminul Haque:** is a software engineer with full-stack development experience. His technical skills include JavaScript, TypeScript, C#, Python, Node.js, Nest.js, Express.js, ASP.NET, React.js, Next.js, Redux Toolkit, Tailwind CSS, MongoDB, PostgreSQL, MySQL, RESTful APIs, AWS S3, and system design.
4. **Pius Kalambe:** Is an experienced ICT and digital health professional with more than 13 years of experience in health-sector technology. His skills include digital health systems, ICT governance, health information systems support, systems analysis, project management, cybersecurity governance, stakeholder engagement, and leadership.
5. **Almas Ashrafi:** Has experience as a data science developer, with skills in Python, Java, GitHub, Django, Docker, Microsoft Word, and Microsoft Excel. His experience includes data extraction systems, document classification models, LDAP authentication, data modification tools, and server environment management.

2.3 Plan and Time Line

The project will be completed over four months, starting in July, with the goal of preparing the first full draft by the end of October. Work will be divided across literature review, data collection, preprocessing, model development, evaluation, and report writing. Each team member will have a primary responsibility, but all members will contribute to review, testing, and final editing.

3 Outcomes

Deliverables:

- A technical report (or publication) detailing the methodology and findings.
- An open-source codebase with data processing, model training, and evaluation code.
- Processed datasets (if licensable) for replicability.
- A set of trained models and baseline comparisons.
- Possibly a lightweight demo or dashboard showing predictions.

Month	Main Tasks	Responsibility
July	Finalize research questions, review related work, confirm datasets, define methodology, and set up the project repository.	Nguyen Gia Long, all members
August	Collect cryptocurrency price data, Twitter/X, Reddit, and news sentiment data. Clean raw data and document API/data-source limitations.	Manoj Kumar, Pius Kalambe
September	Engineer price and sentiment features, align time-series data, implement baseline models, and begin LSTM/GRU model development.	Md Aminul Haque, Pius Kalambe
October	Train and evaluate LSTM variants, compare results with baselines, prepare visualizations, write analysis, and complete the first draft.	Almas Ashrafi, all members

Contributions: We expect to contribute (1) a systematic comparison of sentiment-enhanced models vs. baselines on multiple cryptocurrencies, (2) novel insights into how different sentiment sources and features affect forecasting accuracy, and (3) reproducible tools and data for the community.

AI Usage Disclosure

Generative AI was used for grammar checking, formatting, and wording refinement only. All project ideas, methodology, and final content were formulated, reviewed, and verified by the project team.

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